

Yu Li

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Education

George Washington University	Washington, D.C.
<i>Ph.D. Candidate in Electrical and Computer Engineering, GPA: 4.0/4.0</i>	<i>2025.08 – 2029.05</i>
Advisor: Prof. Tian Lan 📧; Work with: Prof. Zhengling Qi 📧	<i>(expected)</i>
Wuhan University, Hongyi Honor College	Wuhan, China
<i>B.Eng. in Microelectronics Science and Engineering, GPA: 3.87/4.0</i>	<i>2021.09 – 2025.06</i>

Experience

ByteDance Inc.	San Jose, CA
<i>Machine Learning Scientist Intern, Tiktok Recommendation Core</i>	<i>2026.05 – 2026.08</i>
◦ Worked on production-scale video recommendation at TikTok. ◦ Built LLM-based ranking models for the ranking stage, integrating content signals into ranking features.	

Research Highlights

- **LLM Reasoning:** Developing credit assignment, preference optimization, and context conditioning methods to improve alignment and reasoning in large language models [P1][P3][P4][P5].
- **Agentic RL:** Designing reinforcement learning frameworks for hierarchical skill acquisition and multi-turn policy optimization across single and multi-agent systems [P2][C2][C5].
- **Efficient AI:** Improving training and inference efficiency for large language models through parameter-efficient fine-tuning and long-context KV cache management [C1][C6][C8].
- **Generative Models:** Exploring diffusion-based generative models for personalized content-style synthesis and cross-modal generation [C3][J2].

Preprints & Under Review

P=Preprint

- [P1] **OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning** 📧
Yu Li, Rui Miao, Tian Lan, Zhengling Qi
- [P2] **ARISE: Agent Reasoning with Intrinsic Skill Evolution in Hierarchical Reinforcement Learning** 📧
Yu Li, Rui Miao, Zhengling Qi, Tian Lan
- [P3] **Right Meets Wrong: Bilateral Context Conditioning with Reward-Confidence Correction for GRPO** 📧
Yu Li, Tian Lan, Zhengling Qi
- [P4] **INSPO: Unlocking Intrinsic Self-Reflection for LLM Preference Optimization** 📧
Yu Li, Tian Lan, Zhengling Qi
 AI with Recursive Self-Improvement Workshop at ICLR, 2026
- [P5] **Hindsight Self-Distillation: Demonstrating to the Teacher for On-Policy LLM Reasoning** 📧
Yu Li, Shu Hong, Tian Lan
Under review at EMNLP 2026, OA: 3.33/5

Publications

C=Conference, J=Journal, †=Equal Contribution

I have published **6** papers in top-tier machine learning and computer vision venues (e.g., COLM, ACL, CVPR, AAAI, MM), with **4** of them as **first author**.

- [C1] **MomentKV: Closing the Directional Gap in KV Cache Eviction for Long-Context Inference**
[COLM, 2026](#) 
Yu Li, Binxu Li, Tian Lan
- [C2] **Reason in Chains, Learn in Trees: Self-Rectification and Grafting for Multi-turn Agent Policy Optimization**
[ACL Findings, 2026](#) 
Yu Li, Sizhe Tang, Tian Lan
- [C3] **CRAFT-LoRA: Content-Style Personalization via Rank-Constrained Adaptation and Training-Free Fusion**
[CVPR, 2026](#) 
Yu Li, Yujun Cai, Chi Zhang
- [C4] **KG-SAM: Encoding Structural Constraints into Segment Anything Models via Probabilistic Graphical Models**
[ICASSP \(Oral\), 2026](#) 
Yu Li, Da Chang, Xi Xiao
- [C5] **ACDZero: Graph-Embedding-Based Tree Search for Mastering Automated Cyber Defense**
[INFOCOM Intelligent Cloud Computing and Networking, 2026](#) 
Yu Li[†], Sizhe Tang[†], Rongqian Chen, Fei Xu Yu, Guangyu Jiang, Mahdi Imani, Nathaniel D. Bastian, Tian Lan
- [C6] **Mixed Text Recognition with Efficient Parameter Fine-Tuning and Transformer**
[ICONIP, 2024](#) 
Yu Li[†], Da Chang[†]
- [C7] **Prompted Information Bottlenecks: Rethinking Layer-Wise Information Allocation in Frozen Vision Transformers**
[ACM MM, 2026](#)
Yuqi Li, Xi Xiao, Yunbei Zhang, Lin Zhao, Yu Li, Aiden Zhao, Tianyang Wang, Hao Xu, Yingli Tian
- [C8] **Calibrating and Rotating: A Unified Framework for Weight Conditioning in PEFT**
[AAAI, 2026](#) 
Da Chang, Peng Xue, Yu Li, Yongxiang Liu, Pengxiang Xu, Shixun Zhang
- [J1] **Dual branch SAM-Transformer Fusion Network for Accurate Breast Ultrasound Image Segmentation**
[Medical Physics, 2025](#) 
Yu Li, Jin Huang, Du Wang, Liye Mei, Cheng Lei *et al.*
- [J2] **SfMDiffusion: Self-supervised Monocular Depth Estimation in Endoscopy Based on Diffusion Models**
[International Journal of Computer Assisted Radiology and Surgery, 2025](#) 
Yu Li, Da Chang, Jin Huang, Du Wang, Liye Mei, Cheng Lei *et al.*

Selected Awards & Honors

- **ICML 2026 Gold Reviewer**, top reviewers recognition. *2026*
- **Innova International Exchange Scholarship**, 6 recipients university-wide, ¥70,000. *2024*
- **Innova Excellence Scholarship**, Top 3%, ¥10,000, twice. *2023, 2024*
- **First-Class Scholarship**, Top 5%, ¥3,000, three times. *2022, 2023, 2024*

Academic Services

- **Conference Reviewer:** {NeurIPS, COLM, ACM MM, ICML, ICLR, AAAI, ICASSP}'2026
- **Journal Reviewer:** TMLR, TPAMI, ToN, Neurocomputing